

Per-turn integration pipeline

Algorithm 1 · 7 deterministic steps · No JSON schema · CPU classifier

Require: dialogue history H , utterance u_t , classifier C , teacher LLM T ,
state-action map A , exemplar pool E

Ensure: teacher response r_t , next state s_t

```
1:      s_t = C(format(H, u_t))           // 34-way SocRule state from classifier
2:      if s_t.stage < current_stage(H):  // Enforce monotonic stage progression
3:          s_t = snap_forward(s_t)
4:      a_t = A[s_t]                     // Deterministic SocRule strategy lookup
5:      E_10 = stage_balanced_sample(E, k=10) // 10 exemplars, 2 per stage
6:      r_t = T(H, u_t, a_t, E_10)        // Teacher generates response (no JSON)
7:      return (r_t, s_t)
```

Line 1 replaces the LLM-consultant call · Line 4 replaces JSON-schema routing